QUIZFORGE ACADEMIC ASSESSMENT

GENERAL ENGINEERING APTITUDE & REASONING

Section A: Quantitative & Reasoning

1. If a train traveling at 72 km/h crosses a 180-meter long platform in 15 seconds, what is the length of the train in meters?
(A) 100 m
(B) 120 m
(C) 150 m
(D) 180 m
Ans: (B)
Explanation: Speed = 72 * (5/18) = 20 m/s. Total distance = speed * time = 20 * 15 = 300 m. Train length = 300 - 180 = 120 m.

2. Find the missing number in the series: 3, 7, 15, 31, 63, ?
(A) 95
(B) 115
(C) 127
(D) 128
Ans: (C)
Explanation: Each term follows x_n = 2 * x_{n-1} + 1. So 63 * 2 + 1 = 127.

3. In a class of 60 students, 40 like Mathematics, 35 like Physics, and 20 like both. How many students like neither Mathematics nor Physics?
(A) 5
(B) 10
(C) 15
(D) 20
Ans: (A)
Explanation: n(M union P) = 40 + 35 - 20 = 55. Neither = 60 - 55 = 5 students.

4. Refer to the flow diagram showing process flow rates. If inlet stream A is 50 kg/h and stream B is 30 kg/h, what is the combined exit rate?
(A) 60 kg/h
(B) 70 kg/h
(C) 80 kg/h
(D) 90 kg/h
Ans: (C)
Explanation: Conservation of mass under steady state: Exit = A + B = 50 + 30 = 80 kg/h.

ANSWER KEY & EXPLANATIONS

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1. B
2. C
3. A
4. C